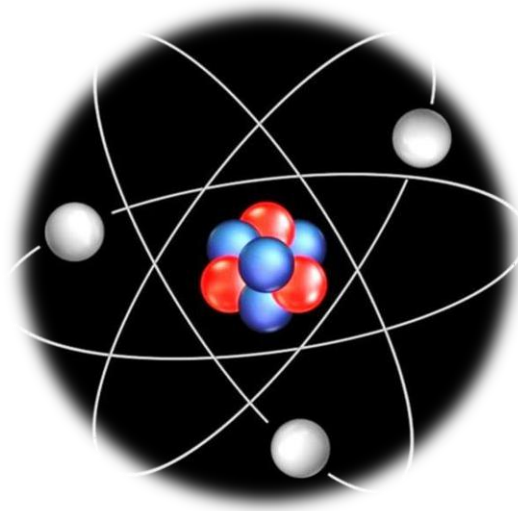


**Department of Science and Humanities
Applied Science Laboratory**



Laboratory Manual

(As per Revised Curriculum SVU R-2025)

FY B Tech AY 2025-26 SEMESTER II

List of Experiments:

Title of the Experiment
Newton's Rings
Hg Spectrum
Hysteresis Effect
Malus law
Resistivity by four probe method

Note: In order to successfully complete the Lab CA in Physics Lab Course (Course code: 316U06L102), students need to perform and prepare a written record of experiments.

Instructions:

1. **Take printouts of only the template (page no 17 and 18) for each experiment.** The template for preparing written record of experiments is given at the end of this manual. You can keep soft copy of the lab manual for reference during the lab sessions in your laptop/mobile phone.
2. Diagrams, observation tables, graphs, calculations etc. should be handwritten. Neatly written and well-organized journal will fetch better grades.
3. Results and/or conclusions should be written for every experiment.
4. Attach graph paper separately. Axes should be labelled. Indicate scale used and units.
5. Attach separate sheet for assignment work.
6. Each experiment should be completed and submitted on time as instructed by the Faculty I/c. Grades will be deducted for late submissions.
7. **Lab attendance should be 100%. Grade penalties will be applied for absenteeism.**
8. Journal carries 30 marks out of 50* marks of Lab CA. Distribution of marks for each experiment will be as follows:

Component	Max marks	Remarks
Performance, written record of experiment and its verification	20	Draw graph on a graph paper and attach with written record of the experiment
Assignments and/or answers to questions asked	05	Write assignment Q&A on separate assignment sheet
Attendance	05	0 for absent, 03 if performed in repetition turn
Total	30	Average of all experiments will be considered for grading the journal

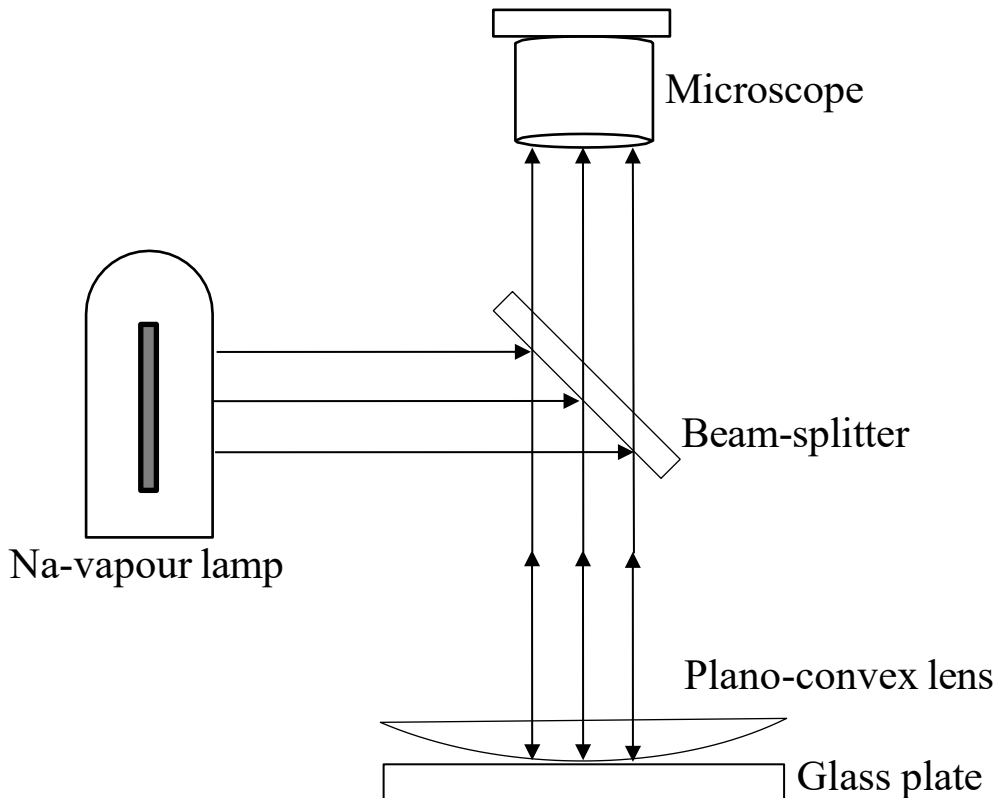
***Remaining 20 marks are allotted to Lab quizzes based on experiments performed**

Title of the experiment: Newton's Rings

Aim: To determine radius of curvature of Plano-convex lens by measuring diameters of interference rings

Apparatus: Newton's rings set-up (pair of plane glass plate and Plano-convex lens, beam-splitter with black box), monochromatic source (Na-vapor lamp), and travelling microscope.

Diagram:



Procedure:

- 1) Arrange apparatus as shown in the diagram. Wait until the Na-vapor lamp turns bright yellow. Observe through the microscope and adjust focus to get a clear Newton's rings interference pattern (alternate dark/bright rings).
- 2) First, adjust crosswire on the center of the pattern. The central spot is taken as $n = 0$ and for the innermost dark ring, $n = 1$. Shift crosswire towards left side of the pattern and count the number of dark rings. Get the crosswire at the 10th dark ring so that the vertical crosswire is tangential to it. Note down the travelling microscope reading at this position.
- 3) Now shift the crosswire towards center of the pattern and adjust it at the 8th dark ring. Follow step 2. Continue this procedure for inner dark rings by skipping one dark ring in-between until you complete reading for 2nd dark ring on the left side of central spot.
- 4) After this, continue shifting the crosswire in same direction (from left to right) so that it moves on the right side of central spot. Adjust the crosswire tangential to 2nd dark ring on right side of

pattern and note down the reading. Continue reading in this manner for outer dark rings by skipping one dark ring in-between until you complete reading for 10th dark ring on the right side of central spot.

- 5) Difference between two readings (i.e. on left and right) for the same ring number will be the diameter (D_n) of that ring. Find diameters of all rings.

Observations:

Sr. No	Ring no (n)	Travelling Microscope Reading (cm)						Diameter D _n (cm) = L~ R [#]	D _n ² (cm ²)
		*On Left L = MSR + VSR (cm)			*On Right L = MSR + VSR (cm)				
		MSR	VSR	L	MSR	VSR	R		
1	12								
--	10								
--	--								
--	--								
6	2								

*MSR: Main scale reading, VSR: Vernier reading = coinciding division x least count,

L: Total reading on left side and R: Total reading on right side of the centre of the ring pattern

[#]The sign ~ means find difference L- R or R- L whichever is positive

Formula: Radius of curvature of lens $R = \frac{\text{slope}}{4\lambda}$

Slope: slope of line D_n^2 v/s n

λ : wavelength of light from Na-vapour lamp = 5890 Å

Graph: Plot a graph of squares of diameters of dark rings as a function of dark ring number. Take D_n^2 along the Y-axis and n along the X-axis. From the points plotted, draw best-fit line that passes through the origin also. Determine slope of this line.

Calculation: Determine radius of curvature of the lens using the formula given.

Result/s and Conclusion:

Radius of curvature of plano-convex lens $R = \underline{\hspace{2cm}}$ cm.

Assignment:

If the entire set-up is immersed in water, diameters of the rings will change.

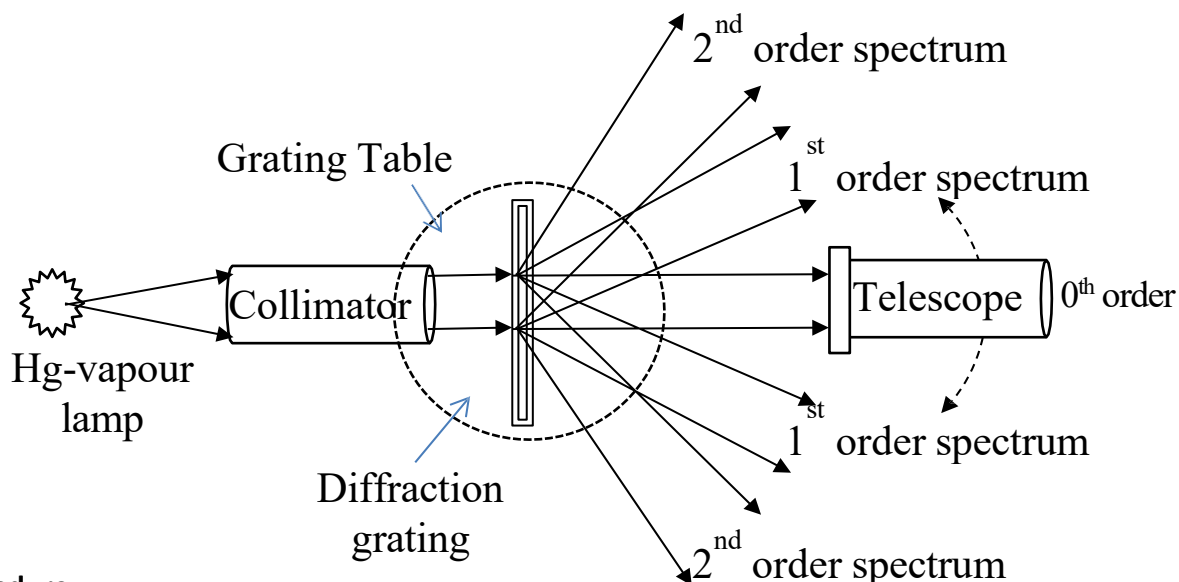
Re-calculate these diameters using the equation: $D_n'^2 = \frac{4nR\lambda}{\mu}$, Where μ is refractive index of water ($\mu = 1.33$) and $D_n'^2$ are diameters of the dark ring in water. Tabulate your answers. Plot a graph of $(D_n')^2$ v/s n in the same graph paper.

Title of the experiment: Hg-Spectrum

Aim: To determine wavelengths of different spectral lines (colors) emitted from a mercury vapour lamp (Hg-source)

Apparatus: Spectrometer, Hg-vapour lamp, and diffraction grating

Diagram:



Procedure:

- 1) Level the spectrometer, prism table, collimator and telescope. Illuminate collimator-slit with Mercury source. Bring telescope in line with collimator and focus it on the illuminated slit. The slit must be sufficiently narrow.
- 2) Adjust the eyepiece of the telescope so that the crosswire is distinctly visible and vertical crosswire is coinciding with the sharp image of the slit. Mount diffraction grating on prism table, perpendicular to incident light (i.e. to the collimator). Lock prism table.
- 3) Move telescope to one side of the incident direction (say, to the left) until you see the first order spectrum. Spectral lines will be visible in the order from violet to red from the incident direction i.e. white line. Focus on the bright-coloured violet/blue spectral line. Adjust the vertical crosswire so that it coincides with the violet/blue line. If required, fix telescope & use its fine motion for this adjustment. Note down readings in both the windows.
- 4) After violet/blue, release the telescope and move it further to get green line. Follow the same procedure as in step 3. Repeat the same for one of the yellow lines and brightest red line from the spectrum.
- 5) Now take the telescope to the right side of the incident direction and follow the procedure of steps 3 and 4.
- 6) The angle 2θ for a particular spectral line is the difference between its readings on the LHS and RHS of incident direction from the same window.

Observations:

Sr. No	Spectral Line	Spectrometer Reading (degrees and minutes)						Diffraction angle # $\theta = \frac{(L \sim R)}{2}$ (deg min)	Sin θ
		*On Left L = MSR + VSR			*On Right R = MSR + VSR				
		MSR	VSR	L	MSR	VSR	R		
1	Blue/Violet								
2	Green								
3	Yellow								
4	Red								

*MSR: Main scale reading, VSR: Vernier reading = coinciding division x least count

L: Total reading on left of the incident direction, R: Total reading on right of the incident direction

The sign ~ means find difference L- R or R- L whichever is positive

Formula: Wavelength of spectral line

$$\lambda = \frac{\sin \theta}{mN} \quad \theta: \text{angle of diffraction maxima}$$

m: order of diffraction maxima. Take m = 1

N: Line density of diffraction grating. N = 5905/cm

Calculation: Calculate wavelengths of spectral lines using the formula given.

Result/s and Conclusion:

Wavelengths of main spectral lines:

Sr. No	Colour	Wavelength (λ in Å)
1	Blue/Violet	
2	Green	
3	Yellow	
4	Red	

Assignment: Determine the highest order of diffraction maxima possible for all the above spectral

lines using $m_{\max} = \frac{1}{N\lambda}$ Note: $m_{\max}$ is an integer value.

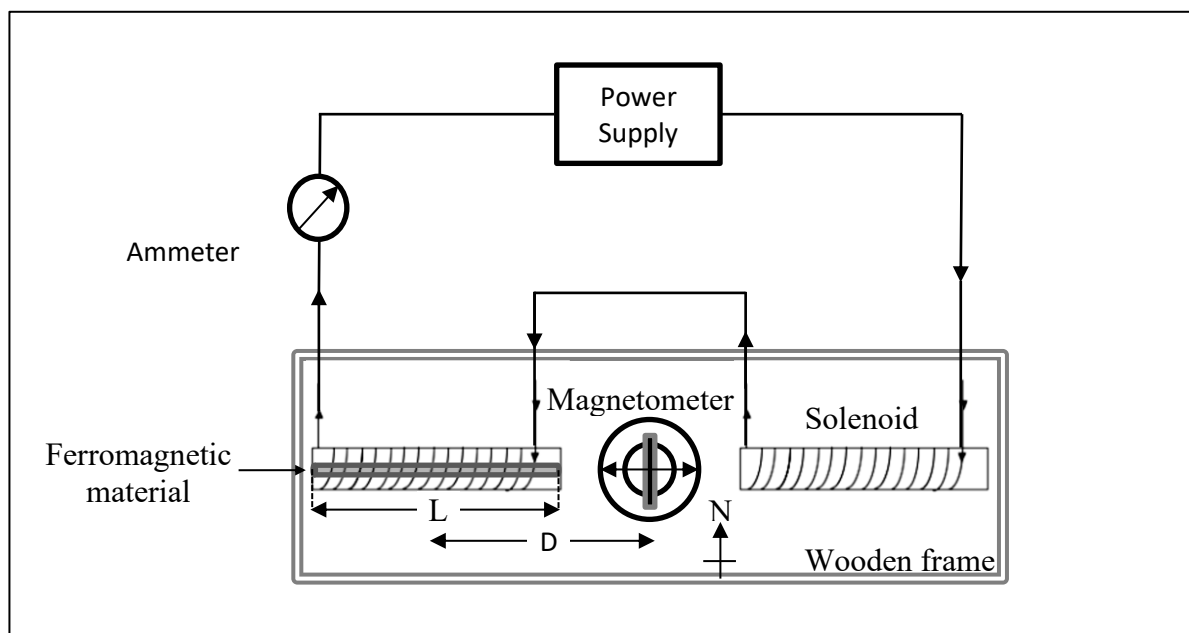
Recalculate the angle of diffraction for the second order using the formula $\theta = \sin^{-1}(mN\lambda)$ for all the spectral lines. Tabulate your answers

Title of the experiment: Hysteresis Effect

Aim: To evaluate coercivity, retentivity and saturation flux density of ferromagnetic sample using hysteresis curve and hence to determine its energy product.

Apparatus: Solenoids, power supply, ferromagnetic sample, magnetometer, ammeter, wooden arms, connecting wires etc

Diagram:



Procedure

- 1) Do not disturb the wooden frame as it is pre-positioned for the experiment. First, disconnect power supply to electromagnet, switch it on, set solenoid current (I_s) to 0 and then connect the power supply back. Check that the magnetometer reads 0-0.
- 2) Increase I_s in positive direction (turn knob clockwise) up to +1.8 A and note down average magnetometer reading (θ).
- 3) Next, start reducing I_s towards 0 (turn knob anticlockwise).
- 4) After $I_s = 0$, continue further to set negative values of I_s up to -1.8 A.
- 5) Now turn the knob clockwise to raise I_s up to 0.
- 6) After $I_s = 0$, continue further to set positive values of I_s up to +1.8 A.
- 7) For steps 2 to 6, follow the steps of I_s as given in observation table and note down θ for each value.

Note:

- 1) If you miss any I_s value, **DO NOT** go back to set that value. Note down magnetometer reading at I_s for next serial number and continue further.
- 2) Whenever magnetometer pointer crosses zero, θ values should be taken positive or negative depending upon the rotation of magnetometer in anticlockwise or clockwise sense.

Observations:

Cycle I: I_s : 0 to +max

Sr. No	1	2	3	4	5	6	7	8	9	10	11
I_s (A)	0	+0.1	+0.2	+0.3	+0.4	+0.5	+0.6	+0.8	+1.0	+1.4	+1.8
Avg θ°											
$\tan(\theta)$											

Cycle II: I_s : +max to 0

Sr. No	12	13	14	15	16	17	18	19	20	21
I_s (A)	+1.4	+1.0	+0.8	+0.6	+0.5	+0.4	+0.3	+0.2	+0.1	0
Avg θ°										
$\tan(\theta)$										

Cycle III: I_s : 0 to -max

Sr. No												
I_s (A)	-0.05	-0.1	-0.15	-0.2	-0.25	-0.3	-0.4	-0.6	-0.8	-1.0	-1.4	-1.8
Avg θ°												
$\tan(\theta)$												

Cycle IV: I_s : -max to 0

Sr. No	35	36	37	38	39	40	41	42	43	44
I_s (A)	-1.4	-1.0	-0.8	-0.6	-0.5	-0.4	-0.3	-0.2	-0.1	0
Avg θ°										
$\tan(\theta)$										

Cycle V: I_s : 0 to +max

Sr. No												
I _s (A)	+0.05	+0.1	+0.15	+0.2	+0.25	+0.3	+0.4	+0.6	+0.8	+1.0	+1.4	+1.8
Avg θ°												
tan (θ)												

Formulae:

$$H_C = N \times X$$

$$B_R = \frac{\pi B_H \left(D^2 - \left(\frac{l}{2} \right)^2 \right)^2 \times Y_R}{\left(\frac{1}{2} \right) \times D \times A}$$

$$B_S = \frac{\pi B_H \left(D^2 - \left(\frac{l}{2} \right)^2 \right)^2 \times Y_S}{\left(\frac{1}{2} \right) \times D \times A}$$

$$E_{\text{loss}} = H_C \times B_R \times V$$

Symbols:

H_C: Coercivity

N: No. of turns per unit length of solenoid

X: Average of the X-intercepts

B_R: Retentivity

B_H: Horizontal component of Earth's magnetic field

D: Distance of solenoid from magnetometer

l: Magnetic length

Y_R: Average of the Y-intercepts

A: Cross-sectional area of sample

B_S: Saturation flux density

Y_S: Average of the saturation Y-values

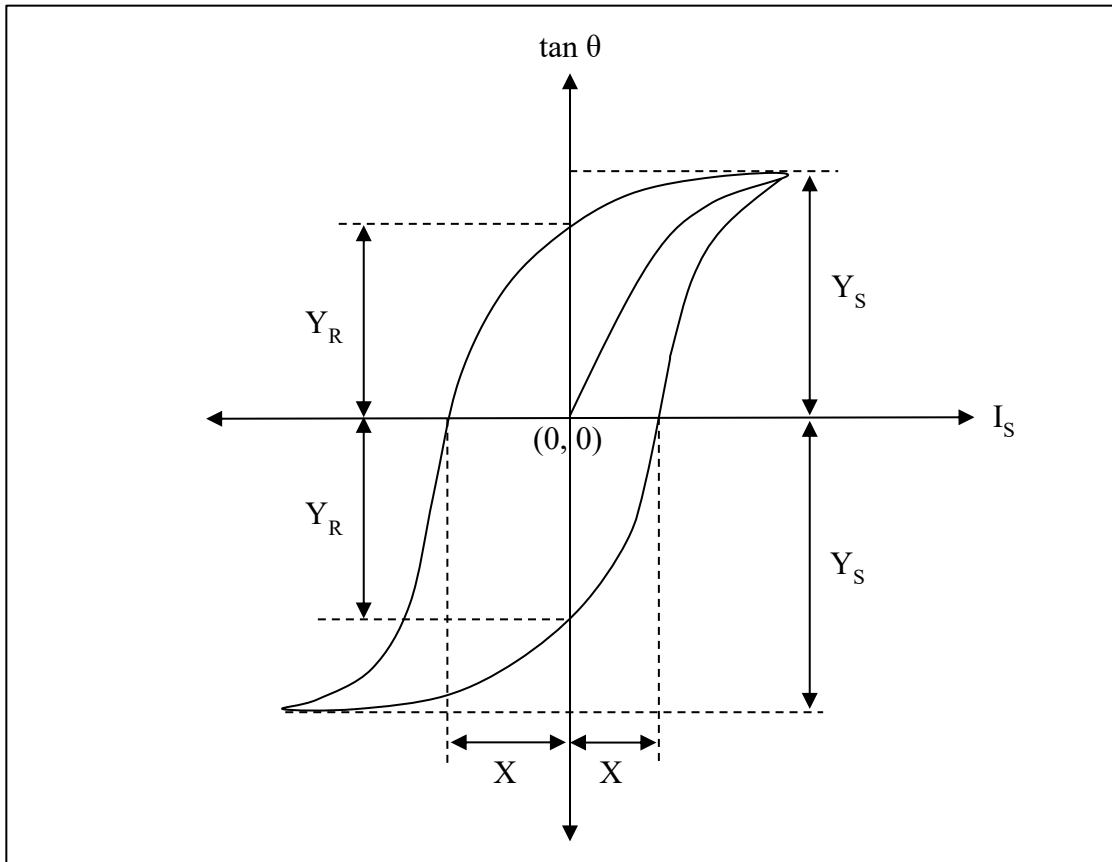
E_{loss}: Energy loss due to hysteresis

V: Volume of sample

Graph:

Plot tan (θ) (Y-axis) v/s I_s (X-axis). It will be a curve advancing towards positive X-axis called hysteresis curve as shown in reference diagram. Mark saturation value of magnetization (tan θ) on the graph. Mark X, Y_R and Y_S from the curve as shown in diagram. Take average of the absolute values of X and Y_R intercepts and saturation Y_S values on both sides to calculate H_C, B_R and B_S respectively. Hence, calculate E_{loss}.

Reference Diagram:



Data:

Horizontal component of Earth's magnetic field (B_H) = $3.81 \times 10^{-5} \text{ Wb/m}^2$

No. of turns of solenoid (N) = 11000/m

Diameter of sample (d) = 6 mm

Length of sample (L) = 10 cm

Data and Evaluated sample parameters:

Horizontal component of Earth's magnetic field B_H	
Distance between magnetometer and solenoid D	
No. of turns of solenoid N	
Diameter of sample d	
Length of sample L	

Magnetic length of sample $l = \frac{5}{6} L$	
Cross sectional area A	
Volume V	

Intercepts and saturation value:

(From plot of $\tan \theta$ vs I_s)

Average X-Intercept#	
Average Y_R -Intercept#	
Saturation Y_S -value#	

Refer to Lab sheet

Results:

Coercivity H_C : _____

Retentivity B_R : _____

Saturation flux density B_S : _____

Energy loss in one cycle Q: _____

Assignment: Complete the following tables (write lower/higher)

Property	Wider hysteresis loop	Narrow hysteresis loop
Permeability		
Retentivity		
Coercivity		
Reluctance		

Also write two applications of ferromagnetic materials having wider hysteresis loop (called hard magnetic materials) and narrow hysteresis loop (called soft magnetic materials).

Title of the experiment: Malus Law

Aim: To determine the relationship between the intensity of the transmitted light through analyzer and ' θ ', the angle between the axis of polarizer and analyzer and to verify Malus Law.

Apparatus: A diode laser, a polarizer-analyzer pair, photo detector, detector output measuring unit (micro ammeter), dial fitted to the polarizer and an optical bench.

Diagram:

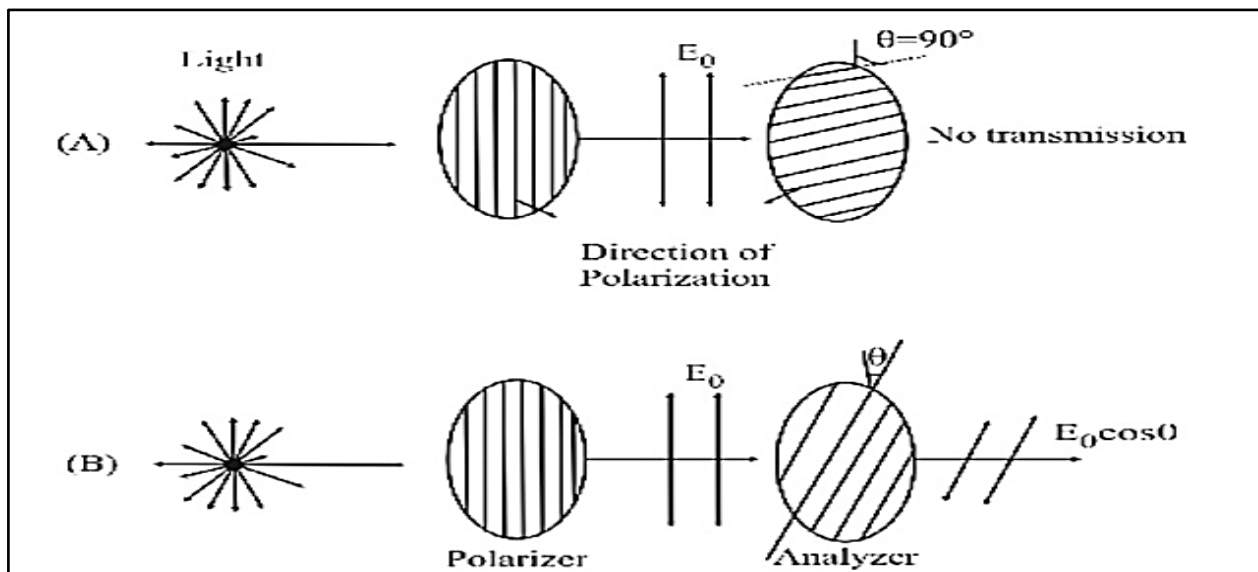


Figure 1

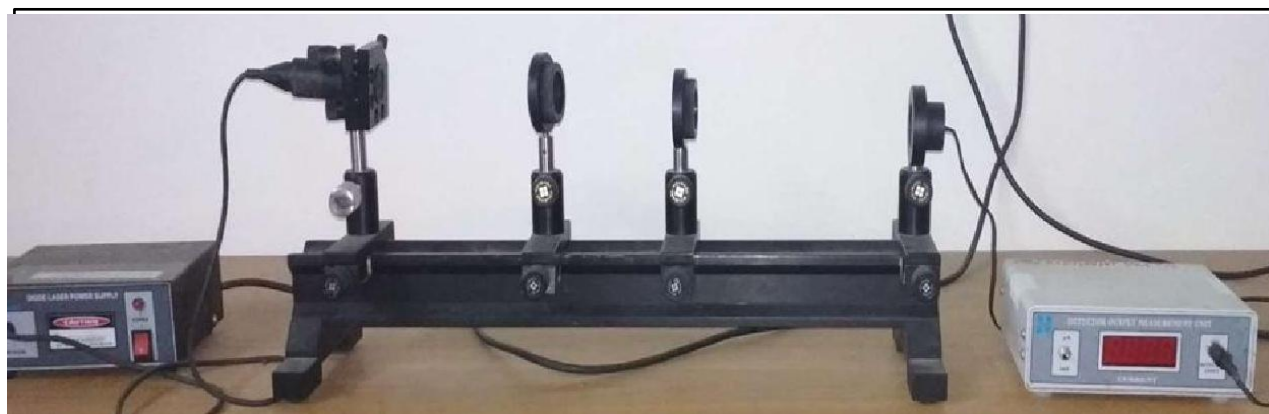


Figure 2

Procedure:

1. Set up the laser, photodiode, the polarizer and analyser as shown in Figure 2.
2. The polarizer and analyser should be normal to the laser beam, and the beam should pass through a "good" portion of the polarizers – look for minimal scattering.
3. When the laser is going through polarizer, analyser and then into a detector then polarizer and analyser transmission axes are parallel.
4. This way you can work with an offset from 0° or any other value. To do this, keep the polarizer fixed and rotate the analyser until you observe a maximum current.
5. Note down the maximum current $I_{\max}$ and angle as φ_0 . At this point the pass axes of polarizer and analyser are parallel.
6. Rotate the analyser in 10° increments from φ_0 and note down this as Φ . Take readings of the intensity at each angle.

Observations:

Angle of Analyzer when current is maximum $\varphi_0 = \dots$ (degrees)

Maximum Current $I_{\max} = \dots$ amp

Sr. No.	Angle of Analyzer Φ (deg)	Angle between the axis of polarizer and Analyzer ' θ ' = $ \varphi - \varphi_0 $ (deg)	$\cos \theta$	$\cos^2 \theta$	Current I_t (amp)
1					
2					
3					
4					
5					
..					
..					
..					
..					
..					

Formula : Intensity of the transmitted light is given by

$$I_t = I_0 \cos^2 \theta$$

Where I_t is the intensity of the light transmitted through the analyzer

I_0 is the intensity of the incident plane polarized light and

θ the angle between the axis of polarizer and analyser

Graph: Plot graph of I_t (on y-axis) against $\cos^2 \theta$ (on x-axis)

Results/ Conclusion: Write on your own how the Malus law have been verified from your experimental data.

Assignment:

An unpolarized light of intensity (I_0) passes through three successive polaroids P_1 , P_2 , and P_3 . The polarization axes of second make an angle of 45° with first, and polarization axes of third make an angle of 30° with second. Determine the intensity of light coming from three polaroids P_1 , P_2 , and P_3 .

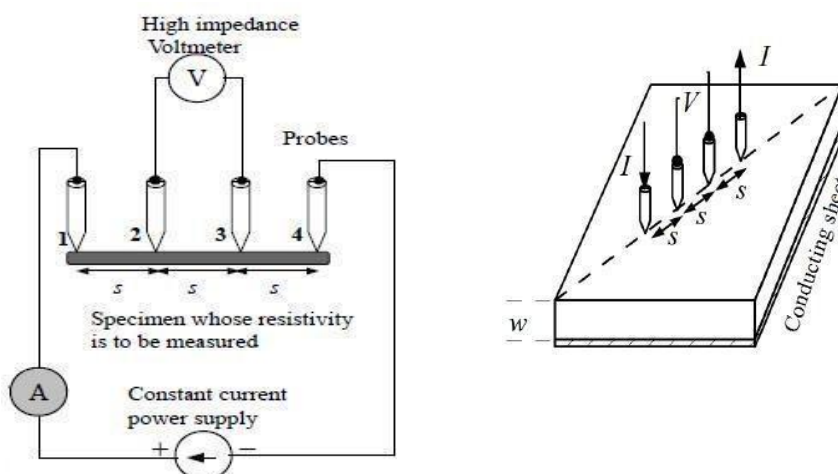
Title of the experiment: Resistivity by 4-Probe Method

Aim: To analyze temperature dependence of resistivity of a semiconductor by four probe method

Apparatus: Probe arrangement, sample, oven 0-200°C, constant current generator, oven power supply and digital panel meter (measuring voltage and current)

url: <https://vlab.amrita.edu/?sub=1&brch=282&sim=1512&cnt=4>

Diagram:



Procedure:

- 1) Select the semiconductor material from the combo box.
- 2) Select the source current from the slider. Restrict the slider based on the range of current.
- 3) Select the Range of oven from the combo box. Set the temperature from the slider.
- 4) Click on the Run Button to start heating the oven in a particular interval, from the default 25°C to the temperature that we set already Click on the Wait button to stop heating.
- 5) Click on the Set button to display the temperature that we set in the oven.
- 6) Click on the Measure button to display the present temperature in the oven.
- 7) Select the range of voltmeter from the combo box. Measure the Voltage using Voltmeter.

Observations:

Material: Germanium

Constant current $I =$ _____ mA

Temperature (T) °C	Temperature (T) K	Voltage (V) mV	Resistivity (ρ) (Ω -m)	$\ln(\rho)$	$\frac{1}{T}$ (K^{-1})
25	25+273 = 298				
30					
--					
--					
70					

Formula: $\rho = 2.13 \times 10^{-3} \frac{V}{I}$

ρ : Resistivity (with geometrical factor)

V: Voltage across the sample

I: Current through the sample

Graph: Plot a graph of resistivity (ρ) as a function of temperature (T). Take ρ along the Y-axis and T along the X-axis.

Result: Resistivity of semiconductor at room temperature (ρ) = _____ $\Omega\text{-m}$.

Resistivity of semiconductor changes ___ with temperature.

Assignment:

Calculate log of resistivity [$\ln(\rho)$] and inverse temperature ($1/T$) using values of resistivity calculated at different temperatures

Temperature (T) °C	Temperature (T) K	Resistivity (ρ) ($\Omega\text{-m}$)	$\ln(\rho)$	$\frac{1}{T}$ (K^{-1})
25	25+273 = 298			
30				
--				
--				
--				
70				

Plot a graph of log of resistivity [$\ln(\rho)$] as a function of inverse temperature ($1/T$). Take $\ln(\rho)$ along the Y-axis and $1/T$ along the X-axis. For the points plotted, draw best-fit line. Determine slope of this line. Calculate energy band gap of semiconductor, using the formula given. Express it in units of eV.

Energy Band Gap $E_g = \kappa \times |\text{slope}|$

|slope| : absolute value of slope of line $\ln \rho$ v/s $\frac{1}{T}$

κ : Boltzmann constant. $\kappa = 1.38 \times 10^{-23} \text{ J/K}$

T: absolute temperature

Energy band gap of semiconductor (E_g) = _____ eV.



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)

Name:
Roll Number:
Batch:
Experiment performed on (date):

Title of the Experiment: Aim:

Apparatus:

Diagram:

Observations:

Calculations:

Result/s and Conclusion/s: